Continuous integration (CI)
- **Objective:** Automate the integration of code changes from multiple contributors into a shared repository.
- **Benefits:** Detecting and resolving integration issues early, ensuring a more stable and reliable codebase.

Continuous deployment (CD)
- **Objective:** Automate the deployment of code changes into production or staging environments.
- **Benefits:** Accelerate the delivery of new features, improvements, and bug fixes with minimal manual intervention.

Model versioning and model registry
Effective model versioning and registry practices are critical in MLOps to maintain transparency, traceability, and ensure the successful deployment and monitoring of machine learning models in production.

	Model versioning	Model registry
Purpose	Track and manage different versions of machine learning models	Centralised repository for storing and managing trained machine learning models
Benefits	Enable reproducibility, facilitate collaboration, and provide a historical record of model changes	Facilitate model discovery, sharing, and deployment across various environments
Implemen-tation	Utilise version control systems (e.g., Git) to track changes in model code, and data configuration	Employ tools like MLflow, TensorBoard, or proprietary solutions to catalogue and organise models in a registry

Components of MLOps
1. Data versioning and management
Data versioning and management are critical components of MLOps that focus on organising, tracking, and controlling changes to datasets throughout the machine learning life cycle.
- **Version control:** Version control systems such as Git are employed to track changes made to datasets, ensuring reproducibility and traceability in experiments.
- **Dataset cataloguing:** A systematic cataloguing system is implemented to manage different versions of datasets, facilitating collaboration among data scientists and maintaining a historical record of dataset changes.
- **Data lineage:** A clear understanding of the data lineage is established, tracking how it evolves. This is crucial for debugging, compliance, and maintaining a transparent data pipeline.

2. Model training and evaluation
Model training and evaluation involve developing and assessing machine learning models.
- **Algorithm selection:** Appropriate algorithms based on the nature of the problem and characteristics of the data are chosen.
- **Feature engineering:** This enhances the model's ability to learn by selecting, transforming, or creating features from the dataset.
- **Hyperparameter tuning:** This optimises the model's hyperparameters to improve its performance.
- **Evaluation metrics:** These define and measure metrics to assess the model's accuracy, precision, recall, and other relevant criteria.

3. Model deployment and monitoring
Model deployment and monitoring are crucial steps in transitioning a trained model into a production environment.
- **Deployment process:** This employs CI/CD pipelines to automate the deployment of models, ensuring a smooth transition from development to production.
- **Scalability:** The deployment process is designed to handle varying workloads and scale resources as needed.
- **Monitoring tools:** Robust monitoring systems are implemented to track the model's performance, detect anomalies, and ensure it operates within specified parameters.
- **Logging:** Logging mechanisms are set up to capture relevant information, facilitating debugging and troubleshooting during deployment.

4. Feedback loops and iterative model improvement
Feedback loops and iterative model improvement establish mechanisms for continuous learning and refinement based on real-world performance.
- **User feedback:** This collects feedback from end users and stakeholders to understand how the model behaves in real-world scenarios.
- **Performance monitoring:** Continuously monitors the model's performance in production, identifying potential issues or opportunities for improvement.
- **Model retraining:** Uses the collected feedback and monitoring data to improve the model iteratively. This may involve retraining with new data or adjusting model parameters.
- **Agile practices:** Implements agile development practices to facilitate quick iterations and adaptations to evolving requirements and insights.